

Module 6 Form

♦ Assignment 1: Short Answer Questions

1. What is an HTML form?

An HTML form is a section of a web page that allows users to enter data and submit it to a server. Forms are used to collect different types of information such as login credentials, registration details, feedback, search queries, and contact information. HTML forms act as a bridge between the user and the backend system by sending user input for processing.

2. What is the purpose of the `<form>` tag?

The `<form>` tag is used to define and group all form-related elements in HTML. It acts as a container for input fields, labels, buttons, dropdowns, and text areas. The `<form>` tag also controls how and where the data is submitted by using attributes like `action` and `method`.

3. What is the use of the `action` attribute in a form?

The `action` attribute specifies the URL or server-side script where the form data will be sent after submission. When the user clicks the submit button, the browser sends the form data to the location defined in the `action` attribute. If the `action` attribute is not specified, the form submits data to the same page by default.

4. What is the use of the `method` attribute?

The `method` attribute defines how form data is sent to the server. The most commonly used methods are:

- **GET:** Sends data through the URL, mainly used for search or non-sensitive data.
- **POST:** Sends data securely in the request body, commonly used for login forms and sensitive information.

Choosing the correct method is important for security and performance.

5. Name any five different `<input>` types.

Some commonly used `<input>` types are:

- `text` – for single-line text input
- `password` – for hidden password input
- `email` – for email address input with validation
- `radio` – for selecting one option from multiple choices
- `checkbox` – for selecting multiple options

6. What is the role of the `<label>` tag?

The `<label>` tag is used to describe an input field. It improves usability and accessibility by allowing users to click on the label text to focus on the associated input field. Labels are especially helpful for screen readers, making forms accessible to visually impaired users.

7. What is the difference between `<input type="text">` and `<textarea>`?

`<input type="text">` is used for short, single-line text input such as names or usernames.

`<textarea>` is used for long, multi-line text input such as comments, feedback, or messages. It allows users to enter multiple lines of text and supports rows and columns for size control.

8. What is the use of the `<select>` tag?

The `<select>` tag is used to create a dropdown list in a form. It allows users to choose one or more options from a predefined list. The options inside a `<select>` tag are defined using `<option>` elements, making it useful for selecting items like country, gender, or category.

9. What is basic form validation?

Basic form validation ensures that the user enters valid and complete information before the form is submitted. It checks conditions such as required fields, correct email format, minimum length, and valid data types. Validation can be done using HTML attributes like **required**, **pattern**, and **type**, or using JavaScript for advanced validation.

10. Why are forms important in real websites?

Forms are essential because they enable interaction between users and websites. They are used for user registration, login systems, feedback forms, online payments, surveys, and data collection. Without forms, websites would be static and unable to collect or process user input, making dynamic web applications impossible.

♦ Assignment 2: Fill in the Blank

1. The tag used to create a form is **<form>**.
2. The attribute that defines where form data is sent is **action**.
3. The input type used to hide text is **password**.
4. The tag used to create a dropdown list is **<select>**.
5. The attribute used to make a field mandatory is **required**.
6. The button that sends form data is called a **submit** button.

♦ Assignment 3: Match the Following

Element/ Attribute	Purpose
<form>	Creates a form and groups form elements
<input>	Used to take user input
<label>	Provides a caption for input fields
<textarea>	Used for multi-line text input
<select>	Creates a dropdown list
required	Makes a form field mandatory

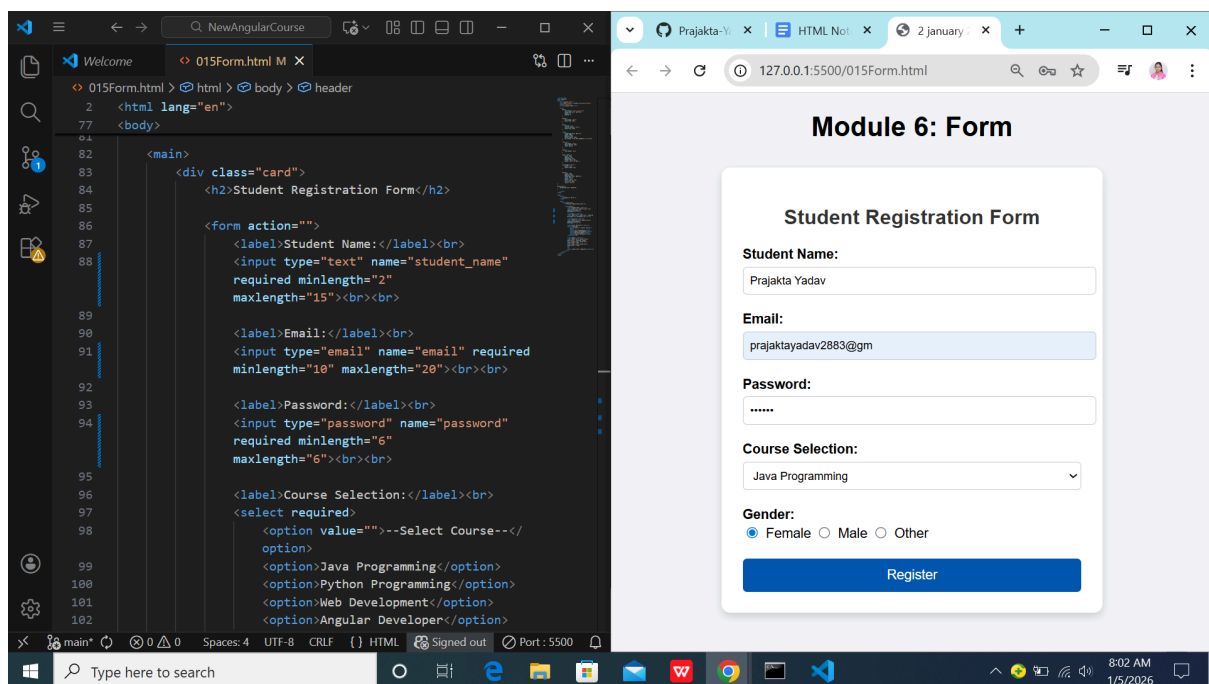
♦ Assignment 4: Identify the Input Type

Write the correct input type for the following fields:

1. **User name** → **text**
2. **Password** → **password**
3. **Email address** → **email**
4. **Mobile number** → **tel**
5. **Date of birth** → **date**

◆ Assignment 5: Practical – Simple Registration Form

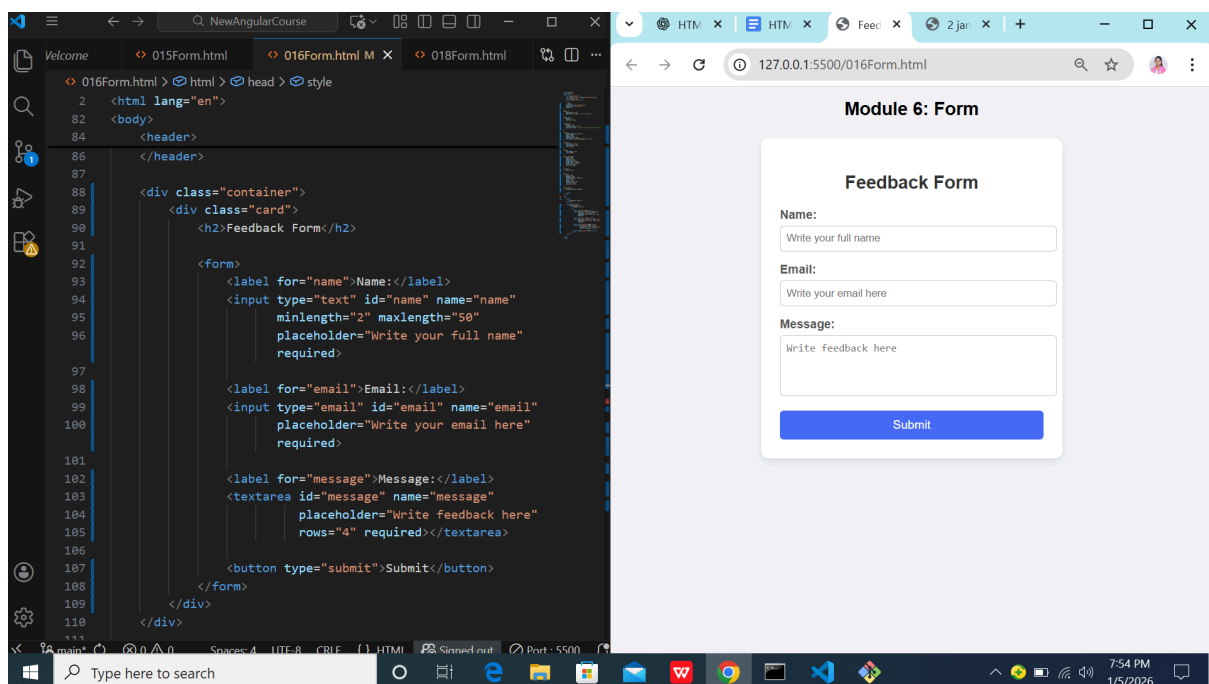
1. Create an HTML form that includes:
2. Student Name (text)
3. Email (email)
4. Password (password)
5. Course selection (dropdown)
6. Gender selection (radio buttons)
7. Submit button



◆ Assignment 6: Practical – Contact / Feedback Form

Create a form that includes:

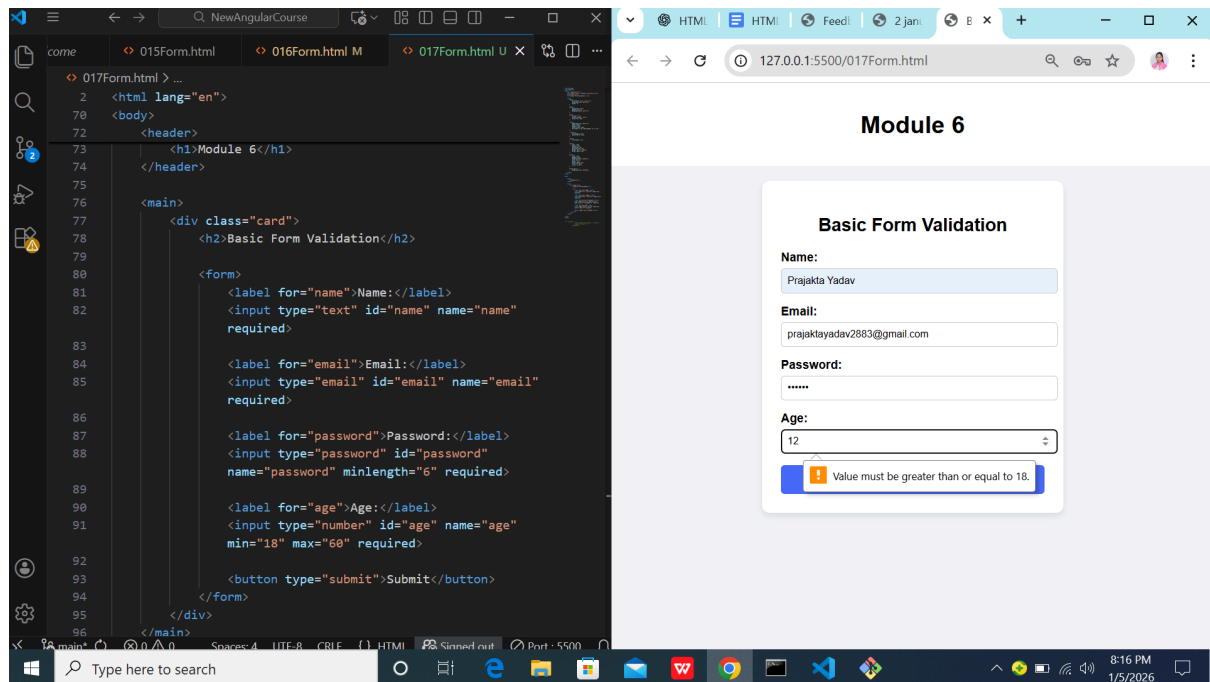
1. Name
2. Email
3. Message (use <textarea>)
4. Submit button



◆ Assignment 7: Basic Form Validation

Create a form with the following validations:

1. Name field should be required
2. Email field should be required
3. Password should have a minimum length of 6 characters
4. Age should be between 18 and 60



◆ Assignment 8: GET vs POST (Conceptual)

Answer the following questions:

1. What is the GET method?

The GET method is used to request or retrieve data from a server.

In this method, form data is visible in the URL as name–value pairs.

2. What is the POST method?

The POST method is used to send data securely to the server.

In this method, form data is sent in the HTTP request body and is not visible in the URL.

3. Which method is more secure and why?

The POST method is more secure because:

- Data is not shown in the URL
- It can send large amounts of data
- It reduces the risk of exposing sensitive information like passwords

4. Which method is suitable for login forms?

The POST method is suitable for login forms because:

- It hides username and password
- It provides better security
- Sensitive data is not stored in browser history

5. Write one real-life use case for GET and POST each.

- **GET Method Use Case:**
Searching for products on an e-commerce website (e.g., search by name or category)
- **POST Method Use Case:**
Submitting a login form or registration form on a website

◆ Assignment 9: Error Identification

Find and correct the errors in the following code:

<form>

<input type="text">

</form>

<input type="submit">

<label>Name</label>

<input type="text" required="yes">

1. Submit button is outside the <form>

- The submit button must be inside the <form> tag to work properly.

2. <label> is not associated with an input

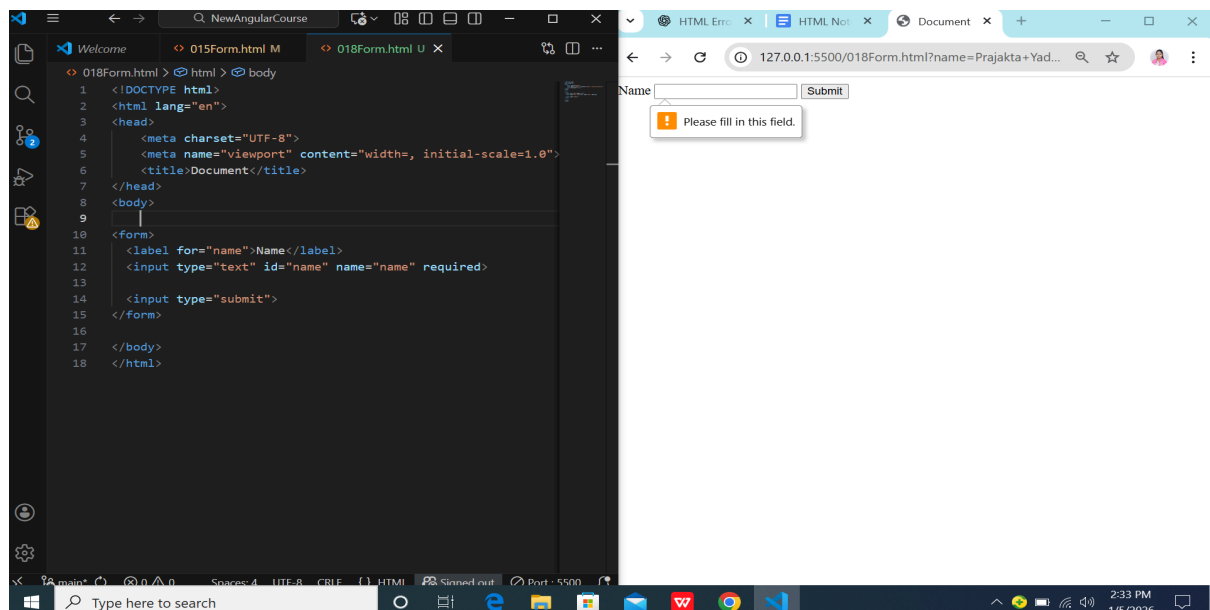
- A label should use the **for** attribute and match an input's **id**.

3. Incorrect use of **required="yes"**

- **required** is a **boolean attribute** and should not have a value.

4. Inputs lack **name** and **id** attributes

- These are important for form handling and accessibility.



♦ Assignment 10: Real-Life Scenario Questions

1. Where are HTML forms used on a college website?

HTML forms are used in many places on a college website, such as:

- Student registration and admission forms
- Login forms for students, teachers, and staff
- Online examination and assignment submission forms
- Feedback and grievance forms
- Library book issue/return request forms
- Contact or inquiry forms

2. Where are forms used on an e-commerce website?

Forms are commonly used on an e-commerce website for:

- User registration and login
- Product search
- Add to cart and checkout process
- Payment and billing details
- Shipping address forms
- Customer reviews and feedback
- Return, refund, and support request forms

3. Why is form validation important?

Form validation is important because:

- It ensures correct and complete data is submitted
- It prevents invalid or empty inputs
- It improves website security
- It provides a better user experience
- It reduces errors in databases
- It helps protect against malicious inputs

4. What problems can occur if validation is not used?

If form validation is not used, the following problems can occur:

- Incorrect or incomplete data submission
- Security risks such as SQL injection or XSS attacks
- System crashes or errors
- Poor user experience
- Increased workload to correct data manually
- Payment and transaction failures